

BTS7960 DC Motor Driver

Specs

Input Voltage: 6-27V

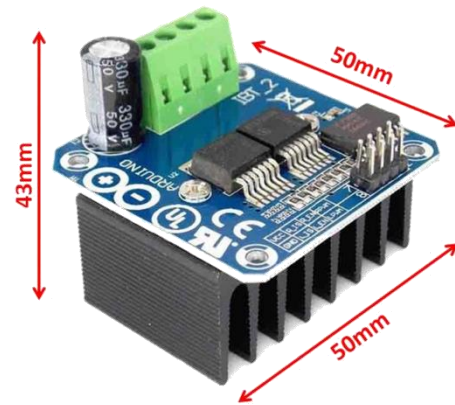
Drive Voltage: 6-27V

Driver Current: 43A MAX Continuous

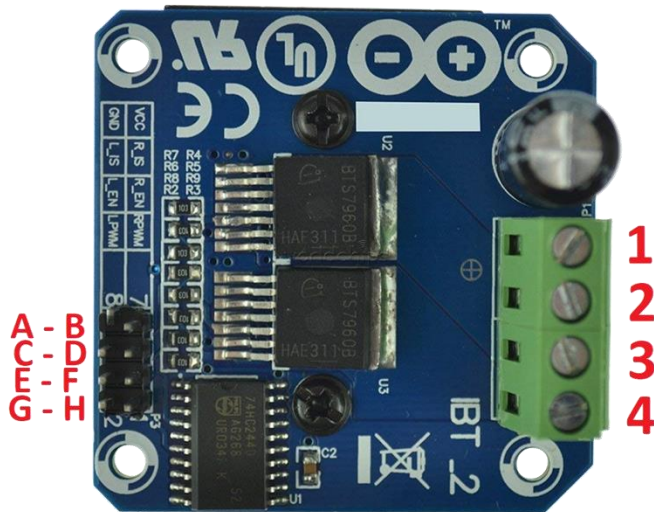
Max Power: 3100 Watts

Logic Input Voltage (VCC): 5V

Logic Voltage: 2-5V On/High, <2V Off/Low



The BTS7960 motor driver will control one motor. It requires a power supply (6-27V DC), a +5V logic power supply, and one to two +5V PWM inputs to control rotational speed and direction. The BTS7960 will operate in three different methods of wiring. The first method is described under Fixed Speed Operation and will allow the motor to operate full speed forward or reverse. Alternately, Methods 2 and 3 are described under Variable Speed Control and allow for motor speed control.



A (GND): ground
B (VCC): controller power supply (5V)
C (L IS): reverse/current alarm output
D (R IS): forward/current alarm output
E (L EN): reverse/enable (active HIGH)
F (R EN): forward/enable (active HIGH)
G (LPWM): reverse/PWM (active HIGH)
H (RPWM): forward/PWM (active HIGH)
1 (B-): motor power supply-
2 (B+): motor power supply+ (5.5-27V)
3 (M+): motor+
4 (M-): motor-

Fixed Speed Operation

- Connect terminal 1 to Ground and terminal 2 to Positive of a 6-27 V power supply
- Connect terminals 3 and 4 to motor positive and negative, respectively
- Connect logic pin A to ground and pins B, E and F to a +5V power supply
- For forward rotation, connect pin H to +5V power supply. For reverse rotation, connect pin G to +5V power supply. This will operate the motor at full speed either forward or reverse

Control method 1 (Full speed, forward or reverse)

PIN H	PIN G	PINS E + F	Motor Output
5 volts	0 volts	5 volts	Full speed Forward
0 volts	5 volts	5 volts	Full speed Reverse

Variable Speed Operation

To control motor speed, you must use a PWM signal. Speed is proportional to % Duty Cycle. You cannot use analog voltage to control speed.

- Connect terminal 1 to Ground and terminal 2 to Positive of a 6-27 V power supply
- Connect terminals 3 and 4 to motor positive and negative, respectively
- Connect logic pin A to ground and pin B +5V power supply
- Connect remaining pins as shown in tables below

Control method 2 (Speed proportional to PWM on PIN H/PIN G. Better for slower speeds but requires two PWM signals, one for Forward and the other for Reverse)

PIN H	PIN G	PINS E + F	Motor Output
PWM (500 Hz, 5V amplitude)	0 volts	5 volts	Variable speed Forward
0 volts	PWM (500 Hz, 5V amplitude)	5 volts	Variable speed Reverse

*At 12V input, motor operates between roughly 15% to 100% duty cycle, 30% to 100% duty cycle under significant load

Control method 3 (Speed proportional to PWM on PINS E + F. Use one PWM signal and send to both E and F. Requires only one PWM signal however there is significant power loss and shorter motor RPM range)

PIN H	PIN G	PINS E + F	Motor Output
5 volts	0 volts	PWM (500 Hz, 5V amplitude)	Variable speed Forward
0 volts	5 volts	PWM (500 Hz, 5V amplitude)	Variable speed Reverse

*At 12V input, motor operates between roughly 55% to 95% duty cycle, 65% to 95% duty cycle under significant load

Troubleshooting

Motor not turning at low speeds or low Duty Cycle

If you are currently controlling the motor using Method 3 above, try using Method 2. Method 2 allows for a lower Duty Cycle before the motor cuts off. If using Method 2 is not possible, try decreasing the voltage supplied to the board so the motor receives a lower voltage at a higher duty cycle.

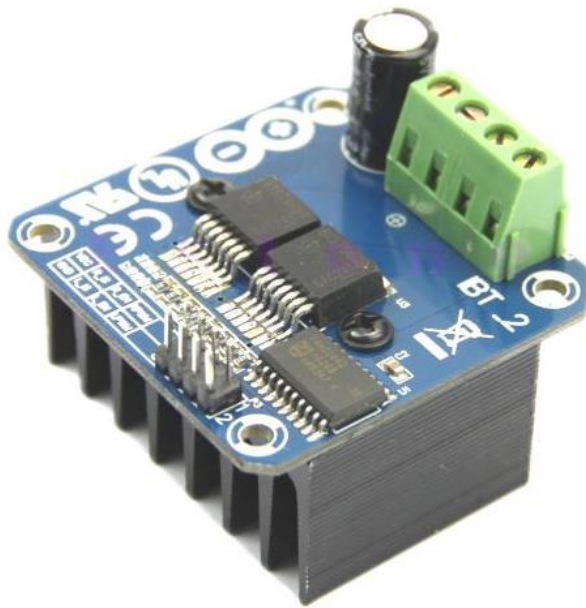
Motor not receiving power

For the motor to receive any power from the motor driver, both the Forward and Reverse Enable pins (pins E and F) must be high (>2V) regardless of whether the motor will spin forward or reverse. If both are high and the motor is still not receiving power, check to make sure +5V is being supplied externally to pin B to power the motor driver board.



BTS7960 High Current 43A H-Bridge Motor Driver

The BTS7960 is a fully integrated high current H bridge module for motor drive applications. Interfacing to a microcontroller is made easy by the integrated driver IC which features logic level inputs, diagnosis with current sense, slew rate adjustment, dead time generation and protection against overtemperature, overvoltage, undervoltage, overcurrent and short circuit. The BTS7960 provides a cost optimized solution for protected high current PWM motor drives with very low board space consumption.

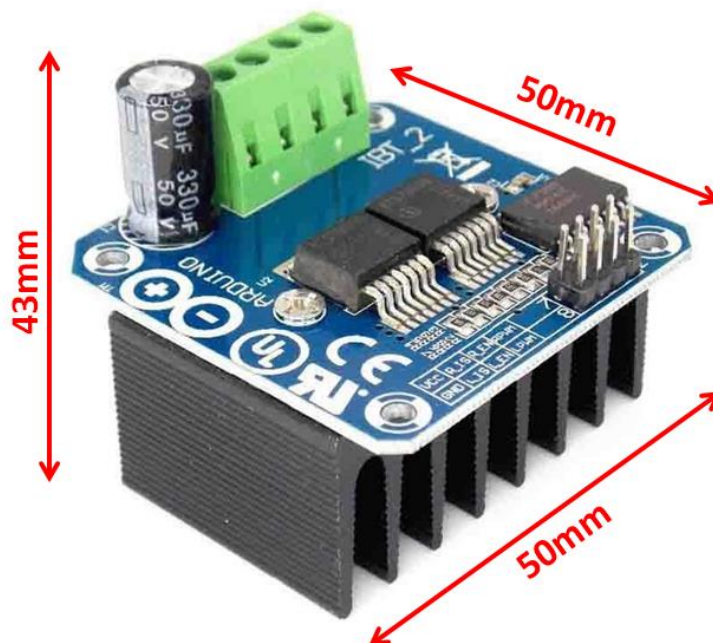
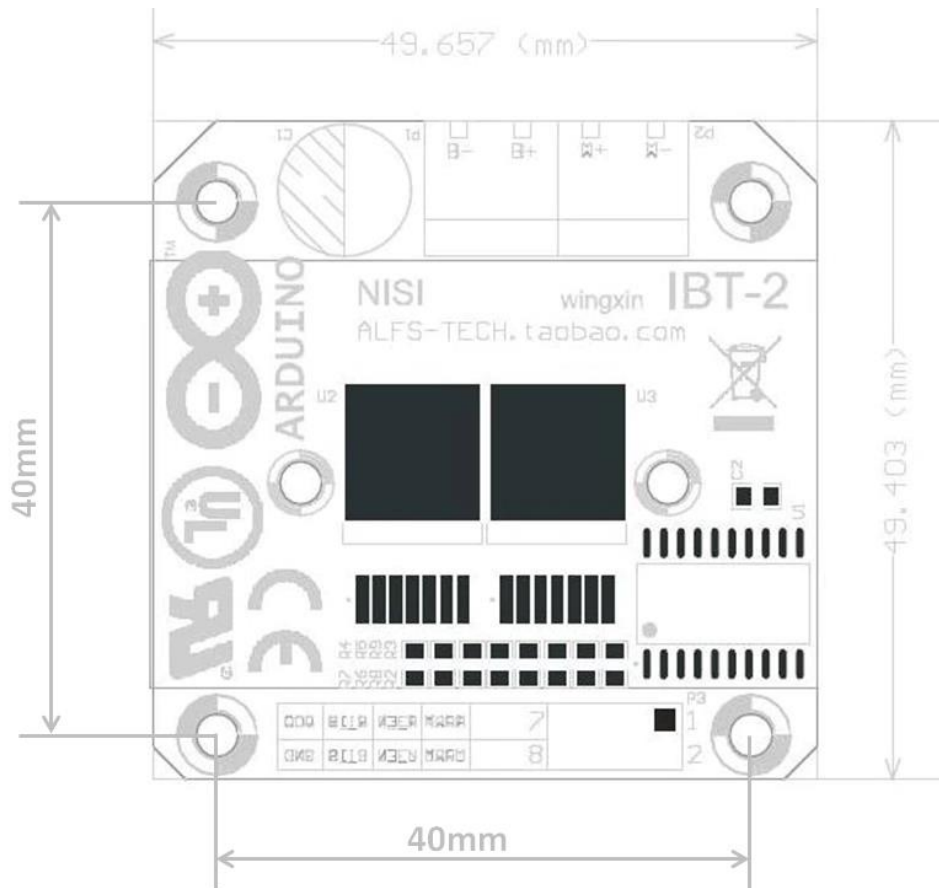


SKU: [DRV-1012](#)

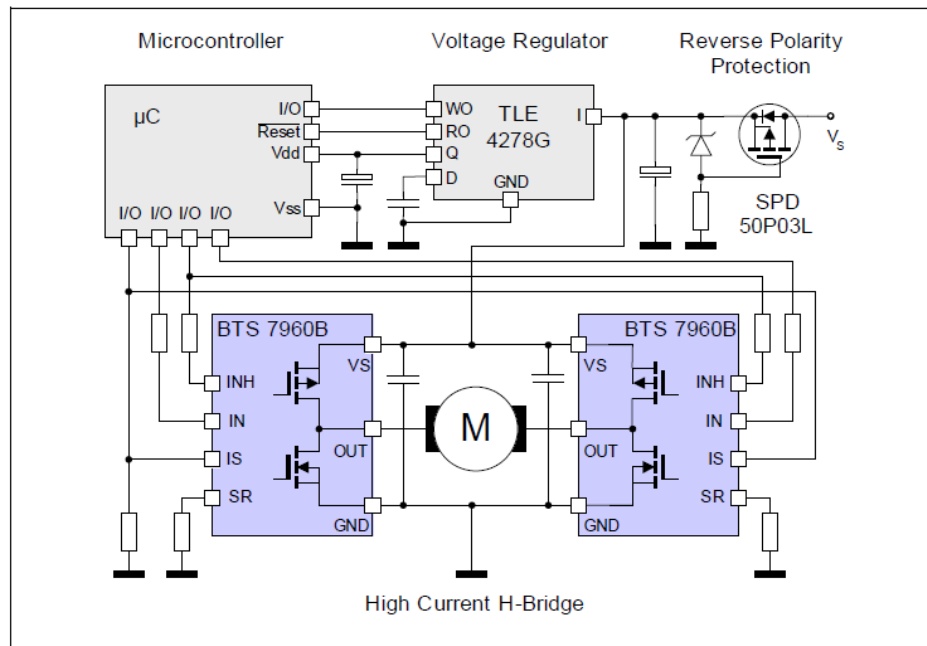
Brief Data:

- Input Voltage: 6 ~ 27Vdc.
- Driver: Dual BTS7960 H Bridge Configuration.
- Peak current: 43-Amp.
- PWM capability of up to 25 kHz.
- Control Input Level: 3.3~5V.
- Control Mode: PWM or level
- Working Duty Cycle: 0 ~100%.
- Over-voltage Lock Out.
- Under-voltage Shut Down.
- Board Size (LxWxH): 50mm x 50mm x 43mm.
- Weight: ~66g.

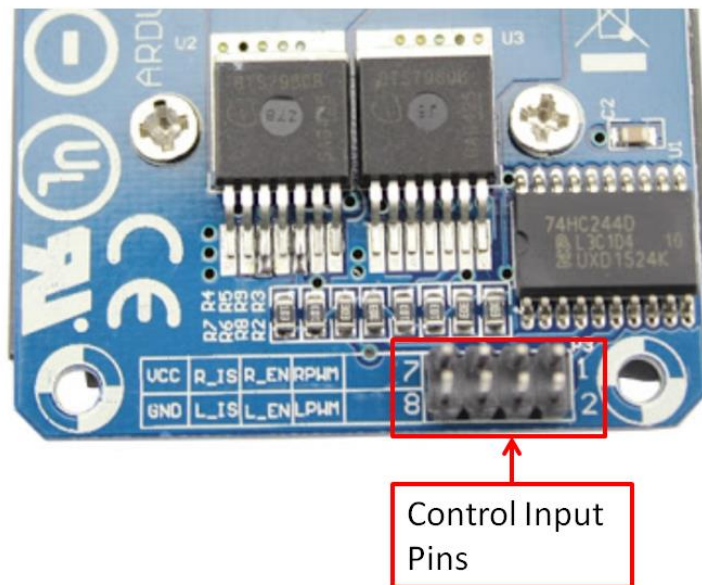
Mechanical Dimension:



Schematic Diagram:

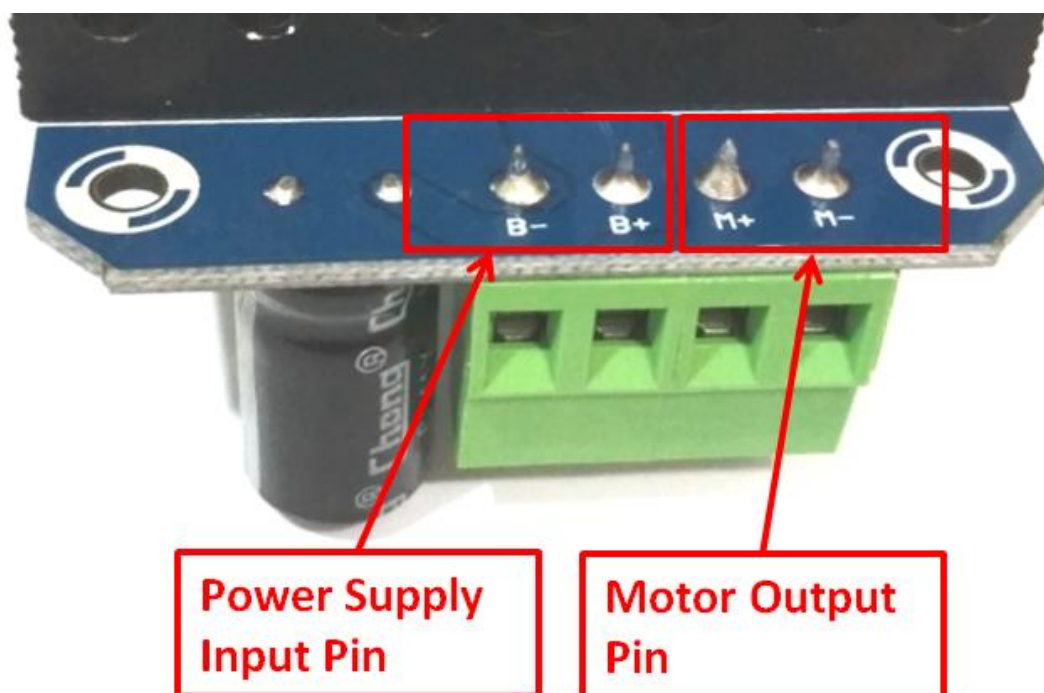


Control Input Pin Function:



Pin No	Function	Description
1	RPWM	Forward Level or PWM signal, Active High
2	LPWM	Reverse Level or PWM signal, Active High
3	R_EN	Forward Drive Enable Input, Active High/ Low Disable
4	L_EN	Reverse Drive Enable Input, Active High/Low Disable
5	R_IS	Forward Drive, Side current alarm output
6	L_IS	Reverse Drive, Side current alarm output
7	Vcc	+5V Power Supply microcontroller
8	Gnd	Ground Power Supply microcontroller

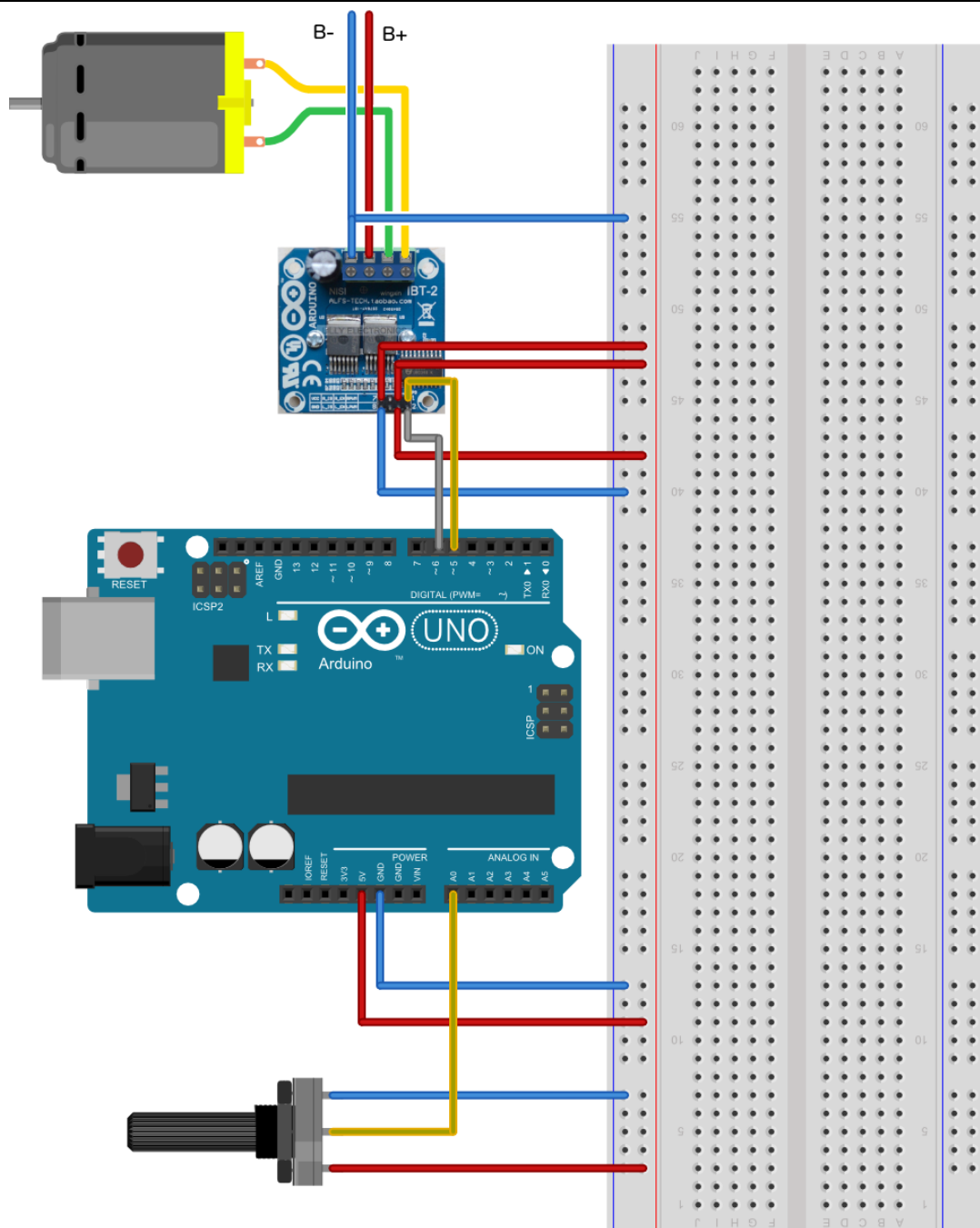
Motor Power Supply & Output Pin Assignment:



Pin No	Function	Description
1	B+	Positive Motor Power Supply. 6 ~ 27VDC
2	B-	Negative Motor Power Supply. Ground
3	M+	Motor Output +
4	M-	Motor Output -

Controlling DC Motor with BTS7960 Using Arduino:

Below is the circuit connection using BTS7960 high power driver to control one DC motor with Arduino board. The potentiometer allows the control of motor speed and rotation direction of the motor.



Arduino to BST7960 Connecting Table:

BST 7960 Pin	Arduino Pin
1 (RPWM)	D5
2 (LPWM)	D6
3 (R_EN)	Arduino 5V
4 (L_EN)	Arduino 5V
5 (R_IS)	Un-connected
6 (L_IS)	Un-connected
7 (VCC)	Arduino 5V
8 (GND)	Arduino GND

Sketch Listing:

Upload the following sketch to Arduino board. Try to turn the potentiometer clock-wise and anti-clock-wise and observe how the motor turn.

```
/*=====
//  Author      : Handson Technology
//  Project     : BTD7960 Motor Control Board driven by Arduino.
//  Description : Speed and direction controlled by a potentiometer attached
//               to analog input A0. One side pin of the potentiometer (either one) to
//               ground; the other side pin to +5V
//  Source-Code : BTS7960.ino
//  Program: Control DC motors using BTS7960 H Bridge Driver.
//=====
// Connection to the BTS7960 board:
// BTS7960 Pin 1 (RPWM) to Arduino pin 5(PWM)
// BTS7960 Pin 2 (LPWM) to Arduino pin 6(PWM)
// BTS7960 Pin 3 (R_EN), 4 (L_EN), 7 (VCC) to Arduino 5V pin
// BTS7960 Pin 8 (GND) to Arduino GND
// BTS7960 Pin 5 (R_IS) and 6 (L_IS) not connected
*/

int SENSOR_PIN = 0; // center pin of the potentiometer

int RPWM_Output = 5; // Arduino PWM output pin 5; connect to IBT-2 pin 1 (RPWM)
int LPWM_Output = 6; // Arduino PWM output pin 6; connect to IBT-2 pin 2 (LPWM)

void setup()
{
    pinMode(RPWM_Output, OUTPUT);
    pinMode(LPWM_Output, OUTPUT);
}

void loop()
{
    int sensorValue = analogRead(SENSOR_PIN);

    // sensor value is in the range 0 to 1023
    // the lower half of it we use for reverse rotation; the upper half for forward
    rotation
    if (sensorValue < 512)
    {
        // reverse rotation
        int reversePWM = -(sensorValue - 511) / 2;
        analogWrite(LPWM_Output, 0);
        analogWrite(RPWM_Output, reversePWM);
    }
    else
    {
        // forward rotation
        int forwardPWM = (sensorValue - 512) / 2;
        analogWrite(LPWM_Output, forwardPWM);
        analogWrite(RPWM_Output, 0);
    }
}
```


Web Resources:

- <https://howtomechatronics.com/tutorials/arduino/arduino-dc-motor-control-tutorial-l298n-pwm-h-bridge/>
- <http://www.labelektronika.com/2016/09/high-current-motor-driver-l298n-arduino.html>



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